

STAAD PRO SYLLABUS

OBJECTIVE: The Objective of the training session is to learn about the Essentials of Structural Design using STAAD PRO to become proficient in the Computer Application to attain the skills required to Design & Analyze the Structure of a building Components. The Training consists of basics Toolsets to create extra ordinary work involving Framed Structure Design , Steel design and Composite Design

Duration :		
S.NO	CONTENT	HOURS (L/P)
1	Introduction to Structural Engineering	
	Structure, Types of Structures	
	Basic Definitions	
	Idealization of Structures	
2	Application Introduction	
	Features, Hardware Requirements	
	STAAD PRO, Screen Organization	
	GUI Overview, Unit Systems	
3	Model Generation	
	Concept of Pre Processor	
	Analysis Engine	
	Post Processor	
	Creating a New File	
	Creating Nodes	
	Adding Beam, Plate, solid, Enhanced Grid Tool (Linear, Radial, Irregular), Geometry Beam Page	
4	Select Menu	
	By all , By Inverse	
	By List	
	By Specification	
5	Model Editing Tools	
	Translational Repeat	
	Circular Repeat	
	Mirror, Rotate	
	Connect Beams Along	
	Stretch Selected Members	
	Intersect Selected Members	
6	Geometry Operations	
	Insert Node: In Existing Member	
	Adding Beams	
	Selecting Members Renumbering	
	For a Single Member	
	For Multiple Members	
	Point to Point	
	Between Midpoints	
	Perpendicular Intersection	
	Curved Member	
	How to Create Beam/ Column & Curved Beams	
	Add Midpoints	

	Add Perpendicular Intersection Beam	
7	Commands	
	Support Specification	
	Member property Specification	
	Member Offset	
	Material Specification	
	Group Specification	
	Release	
	Offset	
	Truss Only	
	Tension Only	
	Compression Only	
8	Material Specification	
	Material Table	
	Modulus Elasticity	
	Weight Density Ratio	
	Poisson's Ratio	
	Co-Efficient of Thermal Expansion, Damping Ratio; Mmember Offset	
9	Load Cases & Analysis	
	Primary Load Menu, Load Commands, Self Weight, Nodal Load	
	Member Load-Concentrated force or moment, Linear Varying, trapezoidal, hydrostatic	
	Perform analysis, run analysis	
	Area Load, Floor Load, Wind Load, Load Combinations, Seismic Definitions	
	Post Analysis Print	
10	Concrete Design	
	Beam Design, Column Design, Design Parameters Selection	
	Defining Parameters, assigning , end Concrete Design	
11	Modeling Plates	
	Geometry-Adding Plate, Create infill Plates, Generate surface meshing, generate Plate mesh, Plate Thickness	
	Pressure on Full Plate, Concentrated Load, Partial Pressure on Plate Load	
	Common Terminologies, Modelling and design procedure	
	Staircase Design	
12	Shear wall Modelling and Design	
	Adding Surface, Commands; Surface thickness, surface load, design parameters	
	Member Load-Concentrated force or moment, Linear Varying, trapezoidal, hydrostatic	
	Perform Analysis, run Analysis	
13	Project	
	Residential Project MultiStori	
	Commercial Project MultiStori	
14	Steel Design	
	Beam Design, Column Design, Design Parameters	
	Truss Design	
	Tower Design	
	Steel Water Tank Design	
	Structure Wizard	
	Quantity Take off	
15	Report generation	
	Footing Design	